A Note for 2301691 : Sheaves and their cohomology

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This is a note taking for the course 2301691 Sheaves and Their Cohomology at CU.

1 Sheaves of Modules

2 Cohomology of Sheaves

3 de Rham Cohomology

3.1 Sheaves of Differential Forms and de Rham Cohomology

We construct the de Rham complex by exterior algebra of the cotangent sheaf $\mathcal{E}_X^1$ and the "differential maps" which is the unique maps satisfy properties we want.

1. There exists a unique family of morphisms $(d^p : \mathcal{E}_X^p \to \mathcal{E}_X^{p+1})_{p\geq 0}$ such that
(1) They are morphisms of sheaves of $\mathbb{R}$-vector spaces,
(2) $d^0 : f \mapsto df$,
(3) For all open subset $U \subseteq openX, \omega \in \mathcal{E}_X^p, \eta \in \mathcal{E}_X^q$, we have $d^{p+q}(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^p \omega \wedge d\eta$,
(4) $d^{p+1} \circ d^p = 0$.
The complex $0 \to \mathcal{O}_X \xrightarrow{d} \mathcal{E}_X^1 \xrightarrow{d^1} \mathcal{E}_X^2 \xrightarrow{d^2} \cdots$ is called the de Rham complex of X.

1.1 Uniqueness : Check locally on charts (U, x). Let $\omega \in \mathcal{E}_X^p(U)$, so $\omega = \sum_{I\in\mathcal{I}} f_I dx^I$ where I is an ordered index set of variable of this chart and dx^I denote the wedge of dx^i over all $i \in I$. We have that $d\omega = \sum_{I\in\mathcal{I}}(df^I \wedge dx^I + (-1)^0 f_I d(dx^I)) = \sum_{I\in\mathcal{I}}(\sum_{i\in I} \partial_i f_I dx^i \wedge dx^I)$.

1.2 Existence : Define d^p locally as above, prove that it satisfies (1)-(4) and use the uniqueness to established that what we are defined agrees on overlapping.
(1) and (2) are obvious.
(3, Exercise) Let $\omega = f_I dx^I \in \mathcal{E}^p$ and $\eta = g_J dx^J \in \mathcal{E}^q$. (The general case can extended by linearlity and distributivity). We have $d(\omega \wedge \eta) = \sum_{i=1}^n \partial_i(f_I g_J) dx^i \wedge dx^I \wedge dx^J$
$= \sum_{i=1}^n (\partial_i(f_I)g_J + \partial_i(g_J)f_I) dx^i \wedge dx^I \wedge dx^J = \sum_{i=1}^n \partial_i(f_I) g_J dx^i \wedge dx^I \wedge dx^J + \sum_{i=1}^n \partial_i(g_J) f_I dx^i \wedge dx^I \wedge dx^J$
$= \sum_{i=1}^n \partial_i(f_I) g_J dx^i \wedge dx^I \wedge dx^J + (-1)^p \sum_{i=1}^n \partial_i(g_J) f_I dx^I \wedge dx^i \wedge dx^J = d\omega \wedge \eta + (-1)^p \omega \wedge \eta$.
(4) $d(d\omega) = \sum_I (\sum_i d(\partial_i f_I dx^i \wedge dx^I)) = \sum_I (\sum_i d(\partial_i f_I) \wedge dx^i \wedge dx^I))$
$= \sum_I (\sum_i (\sum_j (\partial_j \partial_i f dx^j \wedge dx^i \wedge dx^I))) = 0$ (By Schwarz's Theorem).

2. (Poincaré Lemma) $H^i_{dR}(\mathbb{R}^n) = 0$ for all $i > 0$: Obtained by construct a chain homotopy between the identity map and the zero map of de Rham complex $(P^k : \mathcal{E}^k(\mathbb{R}^n) \to \mathcal{E}^{k-1}(\mathbb{R}^n))_{k\geq 0}$, defined by

$$P^k(\omega) = \sum_{I\in\mathcal{I}} \sum_{l=1}^{k} (-1)^{i-1} \left(\int_0^1 t^{k-1} f_I(tx) dt \right) x^{i_l} \quad dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_l}} \wedge \cdots \wedge x^{i_k}.$$

3. The de Rham complex is a Γ-acyclic resolution of $\mathbb{R}_X$: All of $\mathcal{E}^i$ are C^∞-modules and hence Γ-acyclic. The exactness follows from the Poincaré lemma.

4. Functoriality of de Rham cohomology : If $F, G : X \to Y$ are smoothly homotopic, then $H^p_{dR}(F) = H^p_{dR}(G) : H^p_{dR}(Y) \to H^p_{dR}(X)$.

5. If X and Y are homotopy equivalent, then $H^p_{dR}(X) \cong H^p_{dR}(Y)$ for all p. **ask**(If smooth manifolds are homotopy equivalent then is it necessarily that they are smoothly homotopy equivalent?)

Exercise(Arapura 5.2.7-11)

5.2.7. We will say that a manifold is of finite type if it has a finite open cover $\{U_i\}$ such that any nonempty intersection of the U_i are diffeomorphic to the ball. Compact manifolds are known to have finite type [110, pp. 595–596]. Using Mayer–Vietoris and de Rham's theorem, prove that if X is an n-dimensional manifold of finite type, then $H^k(X, \mathbb{R})$ vanishes for $k > n$, and is finite-dimensional otherwise.

<u>*Sol*</u>. (I guess it necessary to required that U_i's are differomorphic to a ball too) Induction on the number of elements of the cover. Base case : $H^{k+1}(U) = H^{k+1}(V) = H^k(U \cap V) = 0$, so both left and right term of $H^k(X)$ vanish, then so is $H^k(X)$.

Induction step : Assume it is true for finite cover of size $< t$. Let X be a finite type wrt $\{U_1, ..., U_{t+1}\}$. Let $Y = \bigcup_{i=1}^t U_i$. Note that $Y \cap U_{t+1} = \bigcup_{i=1}^t (U_i \cap U_{t+1})$, by inductive assumption $H^k(Y \cap U_{t+1}) = 0$ (and also $H^k(Y)$) vanishes for $k > n$. In Mayer-Vietoris we have $\cdots \to H^k(U_{t+1} \cap Y) \to H^{k+1}(X) \to H^{k+1}(U_{t+1}) \oplus H^{k+1}(Y) \to \cdots$. Which implies that $H^p(X)$ vanishes for all $p > n+1$. But what for $p = n+1$?

5.2.8. Let X be a manifold, and let t be the coordinate along $\mathbb{R}$ in $\mathbb{R} \times X$. Consider the maps $i : X \to \mathbb{R} \times X$ and $p : \mathbb{R} \times X \to X$ given by $x \mapsto (0, x)$ and $(t, x) \mapsto x$. Since $p \circ i = id_X$, conclude that $i_* : H^i_{dR}(\mathbb{R} \times X) \to H^i_{dR}(X)$ is surjective.

5.2.9. Continuing the notation from the previous exercise, let $R : E^k(\mathbb{R} \times X) \to \mathcal{E}^k(\mathbb{R} \times X)$ be the operator $(i \circ p)_*$, and let $h : E^k(\mathbb{R} \times X) \to E^{k-1}(\mathbb{R} \times X)$ be the operator that is integration with respect to dt (as in the proof of the Poincaré lemma). Show that $dh + hd = I - R$. Use this to show that R induces the identity on $H^i_{dR}(\mathbb{R} \times X)$. Conclude that $i_* : H^i_{dR}(\mathbb{R} \times X) \to H^i_{dR}(X)$ is also injective, and therefore an isomorphism.

5.2.10. Show that $\mathbb{C} - \{0\}$ is diffeomorphic to $\mathbb{R} \times S^1$, and conclude that $H^1_{dR}(\mathbb{C} - \{0\})$ is one-dimensional. Show that $Re(\frac{dz}{iz}) = \frac{-ydx + xdy}{x^2+u^2}$ generates it.

5.2.11. Let S^n denote the n-dimensional sphere. Use Mayer–Vietoris with respect to the cover $U = S^n - \{\text{north pole}\}$ and $V = S^n - \{\text{south pole}\}$ to compute $H^*(S^n, \mathbb{R})$. (Hint: show that $U \cap V \cong S^{n-1} \times \mathbb{R}$.)

3.2 Compact de Rham and Poincaré duality

In this section, we consider X to be a C^∞ manifold which is orientable ($\exists \omega \in \mathcal{E}^n(X)$ which is nowhere-vanishing).

1. Define compact de Rham complex and integration on n-forms

$$0 \to \mathcal{E}^0_c \xrightarrow{d} \mathcal{E}^1_c \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{E}^n_c \to 0 \qquad \text{and} \qquad \mathcal{E}^n_c(X) \xrightarrow{\int_X} \mathbb{R}.$$

2. For a manifold without boundary X, Stokes' theorem implies that $\mathcal{E}_c^{n-1}(X) \xrightarrow{d} \mathcal{E}_c^n(X) \xrightarrow{\int_X} \mathbb{R}$ is a zero map. Since there is an n-form such that it has nonzero integral, so we have that $H^n_{cdR}(X) \neq 0$. If X is compact then every exact form has zero integral, and thus $H^n_{dR}(X) \neq 0$.

3. The functor Γ_c is a left-exact functor where $\Gamma_c(\mathcal{F}) = \{f \in \Gamma(\mathcal{F}) : \mathrm{supp}(f) \text{ is compact}\}$, and define $H_c^k(X, -)$ to be its right-derived functor.

3.1 Γ_c is a functor : For each $x \in X$ and for each morphism of sheaves $\psi : \mathcal{G} \to \mathcal{G}'$, we have a commutative diagram

$$\begin{tikzcd} \Gamma(\mathcal{G}) \arrow[r, "\Gamma(\psi)"] \arrow[d, "\rho_x"'] & \Gamma(\mathcal{G}') \arrow[d, "\rho'_x"] \\ \mathcal{G}_x \arrow[r, "\psi_x"] & \mathcal{G}'_x \end{tikzcd}$$

which implies that for each $g \in \Gamma(\mathcal{G})$, $g_x = 0$ implies $(\Gamma(\psi)(g))_x = 0$, i.e. $\mathrm{supp}(g) \supseteq \mathrm{supp}(\Gamma(\psi)(g))$.

3.2 Exactness : Let $0 \to \mathcal{F}' \xrightarrow{u'} \mathcal{F} \xrightarrow{u} \mathcal{F}''$ be an exact sequence of sheaves. Note that $0 \to \Gamma_c(\mathcal{F}') \xrightarrow{\Gamma_c(u')} \Gamma_c(\mathcal{F}) \xrightarrow{\Gamma_c(u)} \Gamma_c(\mathcal{F}'')$ is just a subcomplex of the complex given by Γ functor. So $\Gamma_c(u')$ is injective. To prove exactness at $\Gamma_c(\mathcal{F})$, we just nee to prove that for each compact support kernel is an image of some compact support.

4. On paracompact locally-compact Hausdorff space, soft implies Γ_c-acyclic, and thus $H_c^k(X, \mathbb{R})$ can be computed using compact de Rham complex. (see Wedhorn 10.12-16).

5. The first property : $H_c^i(X) \cong H_c^{i+1}(\mathbb{R} \times X)$.

5.1 Construct an explicit chain maps $\pi_* : \mathcal{E}_c^{*+1}(\mathbb{R} \times X) \to \mathcal{E}_c^*(X)$ and $e_* : \mathcal{E}_c^*(X) \to \mathcal{E}_c^{*+1}(\mathbb{R} \times X)$ by

5.2 $\pi_* \circ e_* = id_{\mathcal{E}_c^*(X)}$. which is lifted to $id_{H_c^*(X)}$.

5.3 Construct an explicit chain homotopy $K : \mathcal{E}_c^{*+1}(\mathbb{R} \times X) \to \mathcal{E}_c^*(\mathbb{R} \times X)$ between $id_{\mathcal{E}_c^*(\mathbb{R}\times X)}$ and $e_* \circ \pi_*$ and conclude that its lifted to $id_{H_c^*(\mathbb{R}\times X)}$.

5.4 Hence π_* and e_* are lifted to an isomorphisms of cohomology.

6. Poincaré Lemma for compact support :

$$H^i_{cdR}(\mathbb{R}^n) = \begin{cases} \mathbb{R} & \text{if } i = n, \\ 0 & \text{otherwise.} \end{cases}$$

7. Construct the complex of pseudocurrents

$$\mathcal{D}_X^k(U) = \mathcal{E}_c^{n-k}(U)^\vee = Hom_{\mathbb{R}}()$$

which is a Γ-acyclic resolution of $\underline{\mathbb{R}}$.

8. $H^k(X, \mathbb{R}) \cong \mathcal{H}^i(\mathcal{D}^*(X)) = \mathcal{H}^i(\mathcal{E}_c^{n-*}(X)^\vee) \cong H_c^{n-k}(X, \mathbb{R})^\vee$.

Exercise (Arapura 5.3.7, 5.3.9-10, 5.4.14-16)

5.3.7. Check that $[\alpha] \cup [\beta] = [\alpha \wedge \beta]$ yields a well-defined product on $H^*_{dR}(X)$.

5.3.9. Let $e(X) = \sum (-1)^i \dim H^i(X, \mathcal{R})$ denote the Euler characteristic. Prove that $e(X \times Y) = e(X)e(Y)$.

5.3.10. Show that the cohomology ring of a torus $T = (\mathbb{R}/\mathbb{Z})^n$ is isomorphic to the exterior algebra on $\mathbb{R}^n$.

5.4.14. Prove that the Euler characteristic of an odd-dimensional compact orientable manifold is zero.

5.4.15. If X is a connected oriented n-dimensional manifold, show that

$$H^n(X, \mathbb{R}) \cong \begin{cases} \mathbb{R} & \text{if } X \text{ is compact,} \\ 0 & \text{otherwise.} \end{cases}$$

5.4.16.(a) Show that $\alpha = xdy \wedge dz + ydz \wedge dx + zdx \wedge dy$ generates $H^2_{dR}(S^2)$.

(b) The real projective plane is defined by $\mathbb{RP}^2 = S^2/i$, where $i(x, y, z) = -(x, y, z)$. This is a compact manifold. Show that $H^2(\mathbb{RP}^2) = 0$ by identifying it with the i^*-invariant part of $H^2_{dR}(S^2)$, and conclude that it cannot be orientable.

4 Čech Cohomology

4.1 Čech Complex

In this section we define Čech cochain with respected to a fixed linearly ordered open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of X and a sheaf $\mathcal{F}$.

1. The group of n-cochain

$$C^n = C^n(\mathcal{U}, \mathcal{F}) = \prod_{i_0 < \ldots < i_n} \mathcal{F}(U_{i_0,\ldots,i_n})$$

where $U_{i_0,\ldots,i_n} = U_0 \cap \cdots \cap U_n$ and $\partial^n : C^n \to C^{n+1}$ is defined by $[\partial f]_{i_0,..,i_{n+1}} = \sum_k (-1)^k [f]_{i_0,..,\hat{i_k},..,i_{n+1}}|_{U_{i_0,\ldots,i_{n+1}}}$ where $[g]_{j_0,..,j_t}$ is denote the $\mathcal{F}(U_{j_0,..,j_t})$-component of g. Define $\check{H}^k(\mathcal{U}, \mathcal{F})$ to be the k-th cohomology of this complex.

2. (Exercise) $\partial \circ \partial = 0$. We have $[\partial(\partial f)]_{i_0,..,i_{n+2}} = \sum_k (-1)^k [\partial f]_{i_0,..,\hat{i_k},..,i_{n+2}}|_{U_{i_0,\ldots,i_{n+2}}}$ $= \sum_k (-1)^k (\sum_l (-1)^l [f]_{i_0,..,\hat{i_k},..,\hat{i_l},..,i_{n+2}}|_{U_{i_0,\ldots,\hat{i_k},\ldots,i_{n+2}}})|_{U_{i_0,\ldots,i_{n+2}}} = 0$. The last equality is from the fact that for any $i_k < i_l$, $[f]_{i_0,..,\hat{i_k},..,\hat{i_l},..,i_{n+2}}$ appears two times with coefficients $(-1)^{k+l}$ and $(-1)^{k+l-1}$.

3. Examples :

$$\check{H}^0(\mathcal{U}, \mathcal{F}) = \{(f_i)_{i \in I} \in \prod \mathcal{F}(U_i) : \forall i < j, f_i|_{U_i \cap U_j} - f_j|_{U_i \cap U_j} = 0\} \cong \mathcal{F}(X).$$

$$\check{H}^1(\mathcal{U}, \mathcal{F}) = \frac{\{(f_{ij})_{i<j} \in \prod \mathcal{F}(U_{ij}) : \forall i < j < k, f_{ik} = f_{ij} + f_{jk} \text{ on } U_{ijk}\}}{\{(f_{ij})_{i<j} \in \prod \mathcal{F}(U_{ij}) : \exists (g_i)_{i \in I} \in \prod \mathcal{F}(U_i) \forall i < j, f_{ij} = g_i|_{U_i \cap U_j} - g_j|_{U_i \cap U_j}\}}.$$

4.2 Interpretation of $\check{H}^1$ and $\mathcal{G}$-torsors

For a sheaf of arbitrary group $\mathcal{G}$, we can define $\check{H}^1(\mathcal{U}, \mathcal{G}) = \{(g_{ij}) \in \prod_{i<j} \mathcal{F}(U_{ij}) : g_{ik} = g_{ij} g_{jk} \text{ on } U_{ijk}\}/\sim$. Where $(g_{ij}) \sim (g'_{ij})$ if there is $(r_i) \in \prod \mathcal{F}(U_i)$ such that $g_{ij} = r_i g'_{ij} r_j^{-1}$ on all U_{ij}.

1. For a C^∞- or $\mathbb{C}$-manifold $(X, \mathcal{O}_X)$, there is a 1-1 correspondence

$$\check{H}^1(\mathcal{U}, GL_n(\mathcal{O}_X)) \longleftrightarrow \{\text{rank } n \text{ vector bundles over } X \text{ which is trivial over all } U \in \mathcal{U}\}/\cong .$$

2. Define refine covering, $\check{H}^k(X, \mathcal{F}) = lim_{\leftarrow} H^k(\mathcal{U}, \mathcal{F})$

3. For a C^∞- or $\mathbb{C}$-manifold $(X, \mathcal{O}_X)$, there is a 1-1 correspondence

$$\check{H}^1(\mathcal{U}, GL_n(\mathcal{O}_X)) \longleftrightarrow \{\text{vector bundles of rank } n \text{ over } X\}/\cong \longleftrightarrow \{\text{locally free } \mathcal{O}_X\text{-modules of rank } n\}/\cong .$$

4. For a C^∞- or $\mathbb{C}$-manifold $(X, \mathcal{O}_X)$, define a $\mathcal{G}$-torsor

5. Define the picard group

6. For any space X and a sheaf of groups $\mathcal{G}$ on X, there is a 1-1 correspondence

$$\check{H}^1(X, \mathcal{G}) \longleftrightarrow \{\mathcal{G}\text{-torsors}\}/\cong .$$

7. An exact sequence $0 \to \mathcal{G}' \to \mathcal{G} \to \mathcal{G}'' \to 0$ of sheaves of groups on X induces a long exact sequence

$$1 \to \mathcal{G}' \to \mathcal{G} \to \mathcal{G}'' \to \check{H}^1(X, \mathcal{G}') \to \check{H}^1(X, \mathcal{G}) \to \check{H}^1(X, \mathcal{G}'')$$

Exercises

7.3.10. Show that multiplication in $\mathrm{Pic}(X)$ can be interpreted as tensor product of line bundles. Show that the determinant map det: $\mathrm{GL}_n(\mathcal{R}) \to \mathcal{R}^*$ induces a map $\check{H}^1(X, \mathrm{GL}_n(\mathcal{R}) \to \mathrm{Pic}(X)$ corresponding to the operation $\mathcal{M} \mapsto \bigwedge^n \mathcal{M}$ on locally free sheaves.

7.3.11. Given an exact sequence of sheaves $0 \to \mathcal{A} \to \mathcal{B} \to \mathcal{C} \to 0$ on X and $\gamma \in \mathcal{C}(X)$, define $\delta(\gamma)$ as follows. Choose an open cover $\{U_i\}$ so that $\gamma|_{U_i} = \mathrm{im}(\beta_i)$, with $\beta_i \in \mathcal{B}(U_i)$. Let $\delta(\gamma)$ be the class of $\beta_i - \beta_j \in C^1(\{U_i, \mathcal{A}\})$. (a) Show that $\delta(\gamma)$ gives a well-defined element of $\check{H}^1(X, \mathcal{A})$. (b) Show that $\delta(\gamma) = 0$ if and only if γ lies in the image of $\mathcal{B}(X)$

4.3 The morphism $\check{H}^p(X, \mathcal{F}) \to H^p(X, \mathcal{F})$

We define a sheaf version of the Čech complex by

1. There is a natural map $e : \mathcal{F} \to \mathcal{C}^0(\mathcal{U}, \mathcal{F})$ and we have an exact sequence

$$0 \to \mathcal{F} \xrightarrow{e} \mathcal{C}^0(\mathcal{U}, \mathcal{F}) \xrightarrow{d} \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \xrightarrow{d} \mathcal{C}^1(\mathcal{U}, \mathcal{F}) \xrightarrow{d} \cdots.$$

2. This induces a chain map from above complex to an injective resolution of $\mathcal{F}$, and hence induces a map $\check{H}^p(\mathcal{U}, \mathcal{F}) \to H^p(X, \mathcal{F})$.

3. For each covers $\mathcal{U} < \mathcal{V}$, we have a commutative diagram

$$\begin{tikzcd}
 & \check{H}^p(\mathcal{U}, \mathcal{F}) \arrow[dr, "g"] & \\
\check{H}^p(\mathcal{V}, \mathcal{F}) \arrow[ur, "f"] \arrow[rr, "h"] & & \check{H}^p(X, \mathcal{F})
\end{tikzcd}$$

By the universal property of injective limit, we have a map $\check{H}^p(X, \mathcal{F}) \to H^p(X, \mathcal{F})$.

4. The above map $\check{H}^1(X, \mathcal{F}) \to H^1(X, \mathcal{F})$ for $p = 1$ is an isomorphism.

4.4 Leray Covers

An open cover $\mathcal{U} = \{U_i\}_{i\in I}$ is said to be a Leray cover for $\mathcal{F}$ if for any finite subset J of I, we have $H^1(U_J, \mathcal{F}) = H^2(U_J, \mathcal{F}) = \cdots = 0$ where $U_J = \bigcap_{j\in J} U_j$.

1. If $\mathcal{U} = \{U_i\}_{i\in I}$ is a Leray cover for $\mathcal{F}$ then $\check{H}^k(\mathcal{U}, \mathcal{F}) \cong H^k(X, \mathcal{F})$ for all k.

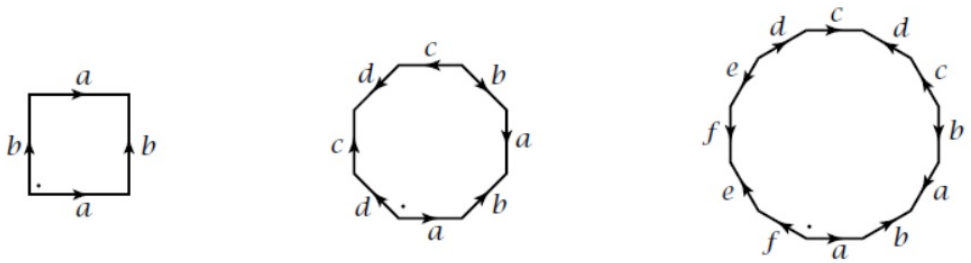
2. One can show that for a paracompact Hausdorff space X, we have $\check{H}^k(X, \mathcal{F}) \cong H^k(X, \mathcal{F})$ for all sheaf $\mathcal{F}$ and all $i \geq 0$.

5 Riemann Surfaces

A Riemann surface X is a 1-dimensional complex manifold. As we know, any n-dimensional complex manifold can be realized as a smooth real manifold of dimension $2n$ equipped with the canonical orientation.

5.1 Σ_g

Denote $\Sigma_0 = S^2$ and for $g \geq 1$ denote Σ_g to be a compact 2-dimensional smooth manifold obtained by identifying sides of $4g$-gon as in the picture (from Hatcher)



1. Fact: Every compact connected orientable 2-dimensional real manifold is homeomorphic to Σ_g for some (and hence exactlty one) g.

2. For each topological space X such that $\forall k \geq 0, dimH^k(X, \mathbb{R}) < \infty$ and $\forall i \gg 0, H^i(X, \mathbb{R}) = 0$ define the Euler Characteristic of X by $e(X) = \sum_{k\geq 0}(-1)^k dimH^k(X, \mathbb{R})$.

3. For an open cover $X = U \cup V$, $e(X) = e(U) + e(V) - e(U \cap V)$.

3.1 Denote $dimH^i(X, \mathbb{R})$ as $h^i(X, \mathbb{R})$.

3.2 Applied the Mayer-Vietoris, give rise to a long exact sequence $\cdots \to H^i(X) \to H^i(U) \oplus H^i(V) \to H^i(U \cap V) \to H^{i+1}(X) \to \cdots$. Applied alternative sum of dimension formula for finite exact sequence, yields $h^0(X) - (h^0(U) + h^0(V)) + h^0(U \cap V) - h^1(X) + (h^1(U) + h^1(V)) - h^1(U \cap V) + \cdots$ which is exactly what we want.

4. $e(\Sigma_g) = 2 - 2g$ and $h^1(\Sigma_g) = 2g$.

5. Generators of $H^1(\Sigma_g, \mathbb{R})$: Fundamental class of curves. For linearly independence : use the intersection pairing (in general for manifold of dimension $2k$, which is the composition of cup product follows by the integration.

6. Example1 : $V_+(f) \subseteq \mathbb{P}^2$ where f is homogeneous of degree d. This is a Riemann surface of genus $\frac{(d-1)(d-2)}{2}$

7. Example2 : Hyperelliptic curve given by the closure of $V(y^2 = \prod_{j=1}^{2g+2}(x - a_j)) \subseteq \mathbb{A}^2$ in $\mathbb{P}^2$. This is a compact Riemann surface of genus g.

5.2 $\bar{\partial}$-Cohomology

Given a smooth function $f : U \subseteq \mathbb{C} \to \mathbb{C}$ in real variables x and y (real and imaginary components). Define

$$\frac{\partial f}{\partial z} = \frac{1}{2}\left(\frac{\partial f}{\partial x} - i\frac{\partial f}{\partial y}\right) \quad \text{and} \quad \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}\left(\frac{\partial f}{\partial x} + i\frac{\partial f}{\partial y}\right).$$

2. By Poincare lemma for $\bar{\partial}$ we have exact sequences

$$0 \to \mathcal{O}_X \hookrightarrow C_X^\infty \xrightarrow{\bar{\partial}} \mathcal{E}_X^{(0,1)} \to 0 \quad \text{and} \quad 0 \to \Omega_X^1 \hookrightarrow \mathcal{E}_X^{(1,0)} \xrightarrow{\bar{\partial}} \mathcal{E}_X^{(1,1)} \to 0.$$

They are also Γ-acyclic resolution since they are C_X^∞-modules (and hence soft).

3.

$$H^1(X, \mathcal{O}_X) \cong \frac{\mathcal{E}_X^{(0,1)}}{\bar{\partial} C_X^\infty} \quad , \quad H^1(X, \Omega_X^1) \cong \frac{\mathcal{E}_X^{(1,1)}}{\bar{\partial}\mathcal{E}_X^{(1,0)}}$$

and $H^k(X, \mathcal{O}_X) = H^k(X, \Omega_X^1) = 0$ for all $k > 1$.

5. We also have an exact sequence $0 \to \mathbb{C}_X \xrightarrow{i} \mathcal{O}_X \xrightarrow{d} \Omega_X^1 \to 0$. Note that we can check the exactness on an open disk. The surjectivity of d is obtained by real Poincare lemma together with the fact that if df is holomorphic then $\frac{\partial f}{\partial \bar{z}} = 0$ and thus f is holomorphic. This yields a long exact sequence

$$0 \to H^0(X, \Omega_X^1) \xrightarrow{\delta^0} H^1(X, \mathbb{C}_X) \xrightarrow{i_*^1} H^1(X, \mathcal{O}_X) \xrightarrow{d_*^1} H^1(X, \Omega_X^1) \xrightarrow{\delta^1} H^2(X, \mathbb{C}_X) \to 0.$$

(Note that $H^0(X, \mathbb{C}_X) \to H^0(X, \mathcal{O}_X)$ is an isomorphism, so the next map is a zero map).

6. We can show that $\delta^0 : H^0(X, \Omega_X^1) \to H^1(X, \mathbb{C}_X) \cong H_{dR}^1(X) \otimes \mathbb{C}$ sends a holomorphic 1-form to its class in de Rham cohomology.

7. For X compact and connected, the map δ^0 is injective.

8. In a further Chapter (Arapura 9.1), we will have a proposition say that $h^0(X, \Omega_X^1) = h^1(X, \mathcal{O}_X) = g$, together with the fact that $h^1(X, \mathbb{C}_X) = 2g$ implies that the map i_*^1 is surjective, and hence above exact sequence breakdown into an exact sequences

$$0 \to H^0(X, \Omega_X^1) \xrightarrow{\delta^0} H^1(X, \mathbb{C}_X) \xrightarrow{i_*^1} H^1(X, \mathcal{O}_X) \to 0$$

and an isomorphism $H^1(X, \Omega_X^1) \cong H^2(X, \mathbb{C}_X) = \mathbb{C}$.

9. Exercise (Arapura 6.2.13) : Show that the differentials $x^i dx/y$, with $0 \leq i < g$, are holomorphic on the hyperelliptic curve. Conclude directly that $h^0(X, \Omega_X^1) = g$ in this case.

5.3 Compact Riemann Surfaces, Meromorphic functions and Divisors

A meromorphic function on X is a holomorphic function. In this section we restrict our attention to the class of **compact** Riemann surfaces.

1. A *divisor* on X is an element of the abelian group

$$\mathrm{Div}(X) = \bigoplus_{x \in X} \mathbb{Z}x$$

and define the degree of $D = \sum n_x x$ by $degD = \sum n_x$. D is said to be *effective* if $\forall x \in X$, $n_x \geq 0$.

2. For a meromorphic function f define $\mathrm{div}(f) = \sum_{x \in X} \mathrm{ord}_x(f)x \in \mathrm{Div}(X)$. The $\mathrm{div}(\omega)$ for a meromorphic 1-form ω is defined similarly. Note that $\mathrm{div} : K(X)^\times \to \mathrm{Div}(X)$ is a group homomorphism.

3. Example : Consider a meromorphic function $z : \mathbb{P}^1(\mathbb{C}) \to \mathbb{C}$ defined by $[1 : a] \mapsto a$.
Around $[0 : 1]$ we have a chart $u : \mathbb{P}^1(\mathbb{C}) \setminus \{[1 : 0]\} \to \mathbb{C}$ defined by $[0 : b] \mapsto b$. Note that on $\mathbb{P}^1(\mathbb{C}) \setminus \{[1 : 0], [0 : 1]\}$, $z = \frac{1}{u}$, and so $dz = \frac{1}{u^2}du$. Hence $\mathrm{ord}_x(dz) = 0$ for every point x except $\mathrm{ord}_{[0:1]}(dz) = 2$.

4. (Riemann-Hurwitz formula) Let $f : X \to Y$ be a non-constant morphism between compact connected Riemann surfaces. We have that $2g_X - 2 = (\deg f)(2g_Y - 2) + \sum_{x \in X}(e_x(f) - 1)$.

4.1 Define the ramification index of f at x, denoted as $e_x(f)$ as follows : Pick a chart u of X around x and a chart v of Y around $f(x)$, their composition with f yields a function on open nbhd of $\mathbb{C}$, then define $e_x(f)$ to be the zero order at $u(x)$ of this function.

4.2 Define the degree of f by $\deg f = \sum_{x \in f^{-1}(y)} e_x(f)$ for some fixed $y \in Y$ (It does not depend on the choice of y, e.g. $z \mapsto z^n$ on $\mathbb{P}^1(\mathbb{C})$, either f has order n or f has order 1 with n preimages).

4.3 To prove the formula, we use triangulations of X and Y in a way that every ramified points are vertices and f is triangle-preserving. After that, it just a combinatorial counting.

5. Let X be a compact Riemann surface of genus g and ω be a meromorphic 1-form on X. We have $\deg(\mathrm{div}(\omega)) = 2g - 2$. Note that it suffice to prove this for a fixed meromorphic 1-form (Every meromorphic forms ω and $\tilde{\omega}$ are differs by some meromorphic function f, i.e. $\tilde{\omega} = f\omega$. Thus we have $\deg(\mathrm{div}(f\omega)) = \deg(\mathrm{div}(f) + \mathrm{div}(\omega)) = \deg(\mathrm{div}(f)) + \deg(\mathrm{div}(\omega)) = \deg(\mathrm{div}(\omega)))$.

6. For each $D \in \mathrm{Div}(X)$, define the sheaf $\mathcal{O}_X(D)$ by

$$\mathcal{O}_X(D)(U) = \{f : U \to \mathbb{C} \cup \{\infty\} : f \text{ is meromorphic and } (f = 0 \text{ or } \forall a \in U, \mathrm{ord}_a(f) + \mathrm{ord}_a(D) \geq 0)\}.$$

7. $\mathcal{O}_X(D)$ is a line bundle and $\mathcal{O}_X(D + D') \cong \mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')$.
Consider a chart $z : U \xrightarrow{\sim}$ open ball $\subseteq \mathbb{C}$. Write $D = \sum_{x \in U} n_x x + E$. Then we have

$$\mathcal{O}_X(D)(U) = \mathcal{O}_X(U) \frac{1}{\prod_{x \in U} (z - x)^{n_x}}.$$

Write $D' = \sum_{x \in U} n'_x x + E'$, we have that

$$[\mathcal{O}_X(D) \otimes_{\mathcal{O}_X} \mathcal{O}_X(D')](U) \cong \mathcal{O}_X(U) \frac{1}{\prod_{x \in U} (z - x)^{n_x + n'_x}}.$$

8. Denote $\omega_X = \Omega^1_X$, $L'(D) = \Gamma(X, \mathcal{O}_X(D))$ and $\ell(D) = dim_{\mathbb{C}} L'(D)$.

9. For a given meromorphic 1-form α on X, we have that $\mathcal{O}_X(K) \cong \omega_X$ where $K = \mathrm{div}(\alpha)$. More generally, by setting $\omega_X(D)(U) = \{\beta : \beta \text{ is a meromorphic 1-from and } \forall x \in U \mathrm{ord}_x(\beta) + \mathrm{ord}_x(D) \geq 0\}$, we have $\omega_X(D) \cong \mathcal{O}_X(K + D)$.

5.4 Riemann-Roch and Projective Embedding

We will prove Riemann-Roch Theorem and use it to show that an Elliptic curve can be embedded to the projective space. We will consider the class of **connected compact** Riemann surfaces.

1. We denote $\underline{A}_x$ to be the skycraper sheaf in x with value in A. We have the following observations:
- $\underline{A}_x$ is flabby and hence $H^k(X, \underline{A}_x) = 0$ for all $k > 0$.
- Since X is a compact Riemann surface, we have an exact sequence

$$0 \to \mathcal{O}_X(-x) \hookrightarrow \mathcal{O}_X \to \underline{\mathbb{C}}_x \to 0$$

and by tensoring along with a line bundle $\mathcal{L} \cong \mathcal{O}_X(D)$, we get

$$0 \to \mathcal{O}_X(D-x) \hookrightarrow \mathcal{O}_X(D) \to \underline{\mathbb{C}}_x \to 0,$$

which give rise to a long exact sequence

$$0 \to H^0(X, \mathcal{O}_X(D-x)) \to H^0(X, \mathcal{O}_X(D)) \to \mathbb{C} \to H^1(X, \mathcal{O}_X(D-x)) \to H^1(X, \mathcal{O}_X(D)) \to 0.$$

2. For any $D \in \mathrm{Div}(X)$, $h^0(X, \mathcal{O}_X(D)), h^1(X, \mathcal{O}_X(D)) < \infty$ and $h^k(X, \mathcal{O}_X(D)) = 0$ for all $k \geq 2$. This can be done by induction on the absolute degree $\sum |n_x|$. For $\sum |n_x| = 0$, we have $D = 0$. Obviously $H^0(X, \mathcal{O}_X) = \mathbb{C}$, So $h^0(X, \mathcal{O}_X) = 1$. For $h^1(X, \mathcal{O}_X)$, consider

5. (Riemann-Roch) For a Riemann surface of genus g with the canonical divisor K,

$$\ell(d) - \ell(K - D) = \deg D + 1 - g_X \qquad \text{for all } D \in \mathrm{Div}(X).$$

From the equation $\ell(d) - \ell(K - D) = \deg D + 1 - h^1(X, \mathcal{O}_X)$, substitude $D = 0$ and $D = K$ yields equations $1 - \ell(K) = 0 + 1 - h^1(X, \mathcal{O}_X)$ and $\ell(K) - 1 = (2g_X - 2) + 1 - h^1(X, \mathcal{O}_X)$. Solving these equation yields $h^1(X, \mathcal{O}_X) = \ell(K) = g_X$.